NONSURJECTIVITY AND COMPLETE FIBERS OF THE SINH–GORDON K-TRANSFORM

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Abstract. The binary-labelled K-transform is a standard device for constructing Sinh–Gordon form factors from entire periodic auxiliary functions. Its converse and its unrestricted fibers have remained unclear. We fix the two diagonal values in the auxiliary annihilation recurrence as parameters and work without a growth condition. First, we disprove universal surjectivity. At coupling $b = 1/4$ and spin zero, Pillin's full pure-A_4 bootstrap family, normalized by $\mathcal{K}_0 = \mathcal{K}_2 = 0$ and $\mathcal{K}_4 = 1$, has no auxiliary lift for any fixed diagonal pair. The obstruction is a four-particle calculation at two intersecting annihilation divisors: all 25 generators of the homogeneous recurrence module have zero regular transform value, and the lower-particle kernel cannot repair it. Second, for every fixed coupling, integer spin, and fixed recurrence parameters, we classify the complete zero-transform kernel. At each finite quotient stalk it is controlled by the explicit finite matrix $C_x = B_x K_x$, where K_x presents the transform syzygies and B_x is the reduced-divisor recurrence quotient. Cartan's theorem B globalizes every admitted step, and the full kernel is the direct sum of its even and odd inverse-limit path spaces. Every nonempty target fiber is an affine translate of this kernel. The nonsurjectivity theorem is specific to $b = 1/4$; no generic-coupling obstruction is asserted.

1. Introduction

The form-factor bootstrap turns factorized scattering data in 1+1-dimensional integrable quantum field theory into matrix elements of local fields. Its analytic foundations go back to the generalized form-factor axioms of Karowski and Weisz, which rest in turn on the factorized S-matrices of Zamolodchikov and Zamolodchikov, and to their subsequent systematic development [16, 8, 3, 15]. For the Sinh–Gordon model, explicit solutions for fundamental fields, scalar operator towers, and general symmetric polynomial families were obtained in [7, 9, 14]. These form factors now enter rigorous work on correlation functions and the reconstruction of quantum-field theoretic properties [10, 11].

A particularly useful construction writes the nonminimal scalar factor as a finite sum over binary labels. Entire, separately periodic functions $p_n(\boldsymbol{\beta} \mid \boldsymbol{\ell})$ satisfying a simple annihilation recurrence are sent by this K-transform to symmetric meromorphic functions $\mathcal{K}_n(\boldsymbol{\beta})$ satisfying the bootstrap residue equation. Babujian–Karowski-type constructions establish the forward implication [1, 2]; Kozlowski gives the precise Sinh–Gordon formulation used below [12, Prop. 2.3]. The paragraph following that proposition leaves open whether all bootstrap solutions arise in this way. The 2025 correlation-function formulation again states only the forward implication, and its hypotheses add a growth bound on p_n that is not imposed here [10, Prop. 2.1]. Free-field methods give one-to-one correspondences in more restrictive generic descendant classes [13, 6], but they do not exhaust the unrestricted entire periodic category considered here.

This paper gives two complementary answers. The first is negative: the universal surjectivity assertion is false. At the self-dual value $b = 1/4$, a pure polynomial component in Pillin's all-level

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solution gives a spin-zero target with

$$\mathcal{K}_0 = \mathcal{K}_2 = 0, \qquad \mathcal{K}_4 = 1. \tag{1}$$

At the four-particle point $(e^{\beta_1}, \dots, e^{\beta_4}) = (-1, 1, -i, i)$, every transform of a homogeneous recurrence correction has zero regular value. Changes of the two-particle lift that remain recurrence-compatible also contribute zero. Thus (1), and hence the full Pillin family extending it, has no lift. A rank calculation shows that this conclusion holds for every fixed choice of the two diagonal recurrence values, not just for the constant choice usually adopted in examples.

The second answer describes all fibers once nonemptiness is known. For integer spin, periodicity and boost covariance descend the problem to the finite quotient

$$Z_n = \big((\mathbb{C}^*)^n/\mathbb{C}^*\big)/S_n. \tag{2}$$

Clearing the transform row gives a finite local coefficient map A_x. Its syzygies form the exact fixed-particle kernel sheaf. Quotienting boundary values by the four-dimensional space of functions of the annihilating pair gives a second finite map B_x. If K_x is any finite presentation of $\ker A_x$, then

$$C_x = B_x K_x \tag{3}$$

tests whether a lower kernel section has a recurrent zero-transform lift. The admissible boundary image is $\operatorname{im} C_x$, and the homogeneous freedom is $K_x(\ker C_x)$. The quotient (2) is Stein, so Cartan's theorem B removes the remaining global one-step obstruction. Taking all compatible steps separately in the two particle-number parities gives the complete kernel; every nonempty fiber is its affine translate.

Four qualifications are structural. The two diagonal recurrence values are fixed parameters: changing them cannot generally be absorbed into a remainder that is independent of all tail labels. The inverse-limit theorem is an integer-spin statement; noninteger spin gives zero source and target spaces by periodicity. The recurrence is imposed on the reduced annihilation divisor, as in the original point-value condition, not on normal jets. Finally, the explicit empty fiber is proved at $b = 1/4$. An exact correction at the same four-particle locus is nonzero for generic coupling, so the proof does not support a generic-b extrapolation.

Section 2 fixes the normalized function spaces and states the main results. Section 3 constructs the fixed-particle transform sheaves. Section 4 proves the inverse-limit fiber theorem. Section 5 builds the all-level pure-A_4 target, and Section 6 proves that its fiber is empty. Section 7 records low-particle checks and the generic-coupling boundary.

2. The normalized transform and main results

Fix

$$0 < b < \frac{1}{2}, \qquad \widehat{b} = \frac{1}{2} - b, \qquad a = \sin(2\pi b). \tag{4}$$

The scattering function is

$$S(\beta) = \frac{\tanh(\beta/2 - i\pi b)}{\tanh(\beta/2 + i\pi b)}. \tag{5}$$

We use the meromorphic continuation of the minimal factor

$$F(\beta) = \exp\left\{-4\int_0^\infty \frac{\sinh(bx)\sinh(\widehat{b}x)\sinh(x/2)}{x\sinh^2 x}\cos\Big(\frac{x}{\pi}(i\pi - \beta)\Big)\,dx\right\}, \tag{6}$$

initially defined for $0 < \operatorname{Im}\beta < 2\pi$.

Definition 2.1 (Scalar target)**.** For $s \in \mathbb{C}$, an admissible scalar family is a sequence $\mathcal{K} = (\mathcal{K}_n)_{n\geq 0}$ such that $\mathcal{K}_n$ is symmetric, separately $2\pi i$-periodic and meromorphic, has only possible simple poles at $\beta_i - \beta_j \in i\pi(1 + 2\mathbb{Z})$, and satisfies

$$\begin{aligned}&\operatorname*{Res}_{\beta_1=\beta_2+i\pi} \mathcal{K}_n(\boldsymbol{\beta}_n)\, d\beta_1 \\ &\quad = \frac{i}{F(i\pi)} \frac{1 - \prod_{r=3}^{n} S(\beta_2 - \beta_r)}{\prod_{r=3}^{n} F(\beta_2 - \beta_r + i\pi)F(\beta_2 - \beta_r)} \mathcal{K}_{n-2}(\beta_3, \dots, \beta_n).\end{aligned} \tag{7}$$

It has spin s if

$$\mathcal{K}_n(\boldsymbol{\beta}_n + \theta \mathbf{1}_n) = e^{s\theta}\mathcal{K}_n(\boldsymbol{\beta}_n) \qquad (\theta \in \mathbb{C}). \tag{8}$$

No growth condition is imposed.

Let $g : \{0, 1\}^2 \to \mathbb{C}$ be symmetric and fixed, with

$$g_{01} = g_{10} = \kappa := -\frac{1}{aF(i\pi)}, \qquad g_{00}, g_{11} \in \mathbb{C}. \tag{9}$$

The last two values are parameters throughout.

Definition 2.2 (Labelled source)**.** An admissible labelled family is a sequence

$$p_n : \mathbb{C}^n \times \{0, 1\}^n \longrightarrow \mathbb{C} \tag{10}$$

whose rapidity dependence is entire and separately $2\pi i$-periodic, which is invariant under simultaneous permutations of the pairs (β_j, ℓ_j), and which obeys the spin covariance

$$p_n(\boldsymbol{\beta}_n + \theta \mathbf{1}_n \mid \boldsymbol{\ell}_n) = e^{s\theta} p_n(\boldsymbol{\beta}_n \mid \boldsymbol{\ell}_n). \tag{11}$$

On $\beta_1 = \beta_2 + i\pi$ it satisfies

$$\begin{aligned}&p_n(\beta_2 + i\pi, \beta_2, \beta_3, \dots, \beta_n \mid \ell_1, \ell_2, \ell_3, \dots, \ell_n) \\ &\quad = g(\ell_1, \ell_2) p_{n-2}(\beta_3, \dots, \beta_n \mid \ell_3, \dots, \ell_n) + h_n(\ell_1, \ell_2 \mid \beta_2, \dots, \beta_n),\end{aligned} \tag{12}$$

where h_n is independent of every tail label $\ell_3, \dots, \ell_n$. Conditions for all other pairs follow by simultaneous permutation symmetry.

Changing g_{00} or g_{11} changes Definition 2.2. Indeed, the difference between two putative choices contributes $(g' - g)(\ell_1, \ell_2)p_{n-2}$ to (12); for a general lower section this depends on the tail labels and cannot be absorbed into h_n. We therefore make no diagonal-gauge identification.

For $|\boldsymbol{\ell}| = \sum_j \ell_j$, define

$$T_n[p_n](\boldsymbol{\beta}_n) = \sum_{\boldsymbol{\ell}\in\{0,1\}^n} (-1)^{|\boldsymbol{\ell}|} \prod_{1\leq i<j\leq n} \left(1 - i\frac{(\ell_i - \ell_j)a}{\sinh(\beta_i - \beta_j)}\right) p_n(\boldsymbol{\beta}_n \mid \boldsymbol{\ell}). \tag{13}$$

The forward theorem states that $T[p] = (T_n[p_n])_{n\geq 0}$ is an admissible scalar family whenever p is admissible [12, Prop. 2.3].

Proposition 2.3 (Noninteger spin)**.** *If $s \notin \mathbb{Z}$, both admissible spaces are zero. Thus the transform is trivially bijective in this branch.*

Proof. Shift all $n \geq 1$ rapidities by $2\pi i$. Separate periodicity leaves either a scalar or a labelled function unchanged, while covariance multiplies it by $e^{2\pi i s} \neq 1$. Hence every positive-particle function vanishes. At $n = 0$, covariance for every $\theta \in \mathbb{C}$ forces the constant to vanish unless $s = 0$. The recurrence does not create a nonzero level from zero data. □

The first main result disproves the universal converse.

Theorem 2.4 (An empty fiber)**.** *Let $b = 1/4$ and $s = 0$. For every fixed pair $(g_{00}, g_{11}) \in \mathbb{C}^2$ with the off-diagonal value (9), there is a full admissible scalar family $\mathcal{K}^{A_4}$ such that*

$$\mathcal{K}_0^{A_4} = \mathcal{K}_2^{A_4} = 0, \qquad \mathcal{K}_4^{A_4} = 1, \tag{14}$$

and

$$T^{-1}(\mathcal{K}^{A_4}) = \varnothing. \tag{15}$$

In particular, the unrestricted fixed-parameter K-transform is not universally surjective.

The second result is valid at every fixed coupling. Its notation is constructed in Sections 3 and 4.

Theorem 2.5 (Complete fixed-parameter fibers)**.** *Fix $0 < b < 1/2$, $s \in \mathbb{Z}$, and one g as in (9). At each quotient stalk x, the transform syzygies and the reduced-divisor recurrence have a finite presentation*

$$A_x, \qquad B_x, \qquad C_x = B_x K_x, \tag{16}$$

where $\operatorname{im} K_x = \ker A_x$. The complete zero-transform family space is the direct sum of the even and odd inverse-limit path spaces defined by these presentations. If a scalar target $\mathcal{K}$ has one admissible lift p^*, then

$$T^{-1}(\mathcal{K}) = p^* + \ker T. \tag{17}$$

This describes all nonempty fibers; Theorem 2.4 supplies an empty one.

3. The fixed-particle coherent transform

We now assume $s \in \mathbb{Z}$. Put $z_j = e^{\beta_j}$ and $X_n = (\mathbb{C}^*)^n$. The two reduced divisor unions relevant to (13) are

$$A_n^- = \bigcup_{i<j} \{z_i + z_j = 0\}, \qquad A_n^+ = \bigcup_{i<j} \{z_i - z_j = 0\}. \tag{18}$$

The first is the allowed annihilation divisor. The second is an apparent equal-rapidity pole of the individual coefficients, because

$$\frac{1}{\sinh(\beta_i - \beta_j)} = \frac{2 z_i z_j}{(z_i - z_j)(z_i + z_j)}. \tag{19}$$

For a label word $\boldsymbol{\ell}$, abbreviate the coefficient in (13) by

$$c_{\boldsymbol{\ell}}(\boldsymbol{\beta}) = (-1)^{|\boldsymbol{\ell}|} \prod_{i<j} \left(1 - i \frac{(\ell_i - \ell_j) a}{\sinh(\beta_i - \beta_j)} \right). \tag{20}$$

Lemma 3.1 (Equal-rapidity cancellation)**.** *For every simultaneously symmetric labelled section p, the sum $\sum_{\boldsymbol{\ell}} c_{\boldsymbol{\ell}} p_{\boldsymbol{\ell}}$ has no pole on A_n^+. The cancellation holds as a meromorphic identity on each whole component, including its intersections with other divisors.*

Proof. On $z_i = z_j$, pair a label word with the word obtained by interchanging ℓ_i and ℓ_j. Their total signs agree. The residue of the (i, j) factor changes sign, while all other factors are exchanged in pairs. Simultaneous symmetry identifies the two boundary values of p. Words with $\ell_i = \ell_j$ have no pole. Thus paired residues cancel as meromorphic functions on the full component, not merely at its generic points. □

Common scaling of $\boldsymbol{z}$ is the boost action. Let

$$Y_n = X_n/\mathbb{C}^* \simeq (\mathbb{C}^*)^{n-1}, \qquad Z_n = Y_n/S_n. \tag{21}$$

The spin character defines a rank-2^n labelled bundle $E_{n,s} \to Y_n$ and a scalar line $L_{n,s} \to Y_n$. The torus Y_n is affine and Stein. Its finite permutation quotient exists as a normal analytic space [5]; equivalently, its invariant Laurent coordinate ring is finitely generated and Z_n is affine. Hence Z_n is Stein.

Write $\pi : Y_n \to Z_n$. Taking the full permutation action, including the spin and divisor linearizations, define the coherent sheaves

$$\mathscr{D}_{n,s} = (\pi_*\mathscr{E}_{n,s})^{S_n}, \qquad \mathscr{M}_{n,s} = \big(\pi_*(\mathscr{L}_{n,s}(A_n^-))\big)^{S_n}. \tag{22}$$

Their global sections are, respectively, the fixed-level labelled functions in Definition 2.2 and symmetric spin-s meromorphic functions with at most simple annihilation poles. Finite-group invariants are exact in characteristic zero. Lemma 3.1 therefore makes (13) a coherent morphism

$$\mathcal{T}_{n,s} : \mathscr{D}_{n,s} \longrightarrow \mathscr{M}_{n,s}. \tag{23}$$

Set

$$\mathscr{I}_{n,s} = \operatorname{im} \mathcal{T}_{n,s}, \qquad \mathscr{N}_{n,s} = \ker \mathcal{T}_{n,s}. \tag{24}$$

The following local description is what makes (24) explicit. Choose an ordered point $x \in Y_n$. Let

$$R = \mathcal{O}_{Y_n,x}, \qquad H_x \subseteq S_n, \qquad R_0 = R^{H_x} \simeq \mathcal{O}_{Z_n,\pi(x)}, \tag{25}$$

where H_x is the finite stabilizer. Let d_- and d_+ be products of reduced local equations for the components of A_n^- and A_n^+ through x. Each

$$b_{\boldsymbol{\ell}} = d_- d_+ c_{\boldsymbol{\ell}} \tag{26}$$

is holomorphic. Lemma 3.1 says that, for an invariant label vector p, $\sum b_{\boldsymbol{\ell}} p_{\boldsymbol{\ell}}$ is divisible by d_+. In a local target frame define

$$A_x(p) = \frac{\sum_{\boldsymbol{\ell}} b_{\boldsymbol{\ell}} p_{\boldsymbol{\ell}}}{d_+}. \tag{27}$$

This is a finite R_0-linear map between the correctly linearized invariant modules.

Proposition 3.2 (Fixed-level image and kernel)**.** *At every quotient stalk,*

$$(\mathscr{I}_{n,s})_{\pi(x)} = \operatorname{im} A_x, \qquad (\mathscr{N}_{n,s})_{\pi(x)} = \ker A_x. \tag{28}$$

Moreover,

$$\operatorname{im} T_n = \Gamma(Z_n, \mathscr{I}_{n,s}), \qquad \ker T_n = \Gamma(Z_n, \mathscr{N}_{n,s}). \tag{29}$$

On the free ordered locus away from A_n^+, one has $H_x = \{1\}$ and $d_+ = 1$. Thus A_x is the row $(d_- c_{\boldsymbol{\ell}})$, which gives the ordinary ideal and syzygy presentation

$$\mathscr{I}_{n,s} \otimes R = \big(d_- c_{\boldsymbol{\ell}} : \boldsymbol{\ell} \in \{0,1\}^n\big), \tag{30}$$

$$\mathscr{N}_{n,s} \otimes R = \operatorname{Syz}_R\big(d_- c_{\boldsymbol{\ell}}\big). \tag{31}$$

Proof. Equations (26)–(27) clear exactly the local target frame $1/d_-$ and remove the apparent equal-rapidity factor. Thus they give the stalk map of (23), including every stabilizer character, and (28) follows.

The exact sequence

$$0 \longrightarrow \mathscr{N}_{n,s} \longrightarrow \mathscr{D}_{n,s} \longrightarrow \mathscr{I}_{n,s} \longrightarrow 0 \tag{32}$$

consists of coherent sheaves on the Stein space Z_n. Cartan's theorem B gives $H^1(Z_n, \mathscr{N}_{n,s}) = 0$ [4]. Taking global sections proves (29). When H_x is trivial and $d_+ = 1$, the same map is the row $(d_- c_{\boldsymbol{\ell}})$, giving (30)–(31). □

Proposition 3.2 does not assert that the image is the whole allowed-pole line. Indeed, the four-particle point used later is a base point of every cleared generator. Retaining the image ideal is essential when the recurrence is imposed.

4. The recurrence quotient and complete fibers

For an unordered pair $e = \{i, j\}$, let $D_e \subset Y_n$ be the reduced annihilation component $z_i + z_j = 0$. In the label space

$$V_n = \mathbb{C}^{\{0,1\}^n}, \tag{33}$$

let

$$W_e = \{f(\boldsymbol{\ell}) : f \text{ depends only on } (\ell_i, \ell_j)\}. \tag{34}$$

This is four-dimensional and is precisely the allowed remainder space in (12). If $\iota_e : D_e \hookrightarrow Y_n$ is the closed inclusion, form the full edge sum upstairs and only then take invariants:

$$\widetilde{\mathscr{R}}_{n,s} = \bigoplus_e (\iota_e)_*\big((\mathscr{E}_{n,s}|_{D_e})/\mathscr{W}_e\big), \qquad \mathscr{R}_{n,s} = \big(\pi_* \widetilde{\mathscr{R}}_{n,s}\big)^{S_n}. \tag{35}$$

This order is important when a stabilizer permutes annihilation components. Restriction followed by the quotient defines

$$\rho_{n,s} : \mathscr{D}_{n,s} \longrightarrow \mathscr{R}_{n,s}. \tag{36}$$

Let $q_{n-2} \in \Gamma(Z_{n-2}, \mathscr{N}_{n-2,s})$. On D_e, delete the two coordinates in e, pull back q_{n-2}, extend it constantly in the two deleted labels, multiply by $g(\ell_i, \ell_j)$, and take its class modulo W_e. These classes assemble to the invariant global section

$$G_{n,s}(q_{n-2}) = \big([g(\ell_i, \ell_j)\, \delta_e^* q_{n-2}]\big)_e \in \Gamma(Z_n, \mathscr{R}_{n,s}). \tag{37}$$

For $n = 2$ in a kernel family, we use the explicit convention $G_{2,s}(q_0) = 0$, since $q_0 = 0$.

Lemma 4.1 (Recurrence as a quotient equality)**.** *A pair (q_{n-2}, q_n) obeys the homogeneous form of (12) for some tail-label-independent remainders if and only if*

$$\rho_{n,s}(q_n) = G_{n,s}(q_{n-2}). \tag{38}$$

This equality is on the full reduced divisor union and therefore includes shared, disjoint, and higher intersections.

Proof. On D_e, equality of the two quotient classes says exactly

$$q_n - g(\ell_i, \ell_j)\delta_e^* q_{n-2} \in W_e, \tag{39}$$

which is (12) with the lower and upper families replaced by their difference. Conversely, every allowed remainder lies in W_e and vanishes in the quotient. Since (35) includes the whole reduced component sheaf, equality retains all intersection germs. It imposes values, not normal jets or a nonreduced thickening. □

Restrict (36) to the transform syzygies and set

$$\mathscr{P}_{n,s} = \ker(\rho_{n,s}|_{\mathscr{N}_{n,s}}), \qquad \mathscr{J}_{n,s} = \operatorname{im}(\rho_{n,s}|_{\mathscr{N}_{n,s}}). \tag{40}$$

Thus

$$0 \longrightarrow \mathscr{P}_{n,s} \longrightarrow \mathscr{N}_{n,s} \xrightarrow{\rho_{n,s}} \mathscr{J}_{n,s} \longrightarrow 0 \tag{41}$$

is an exact sequence of coherent sheaves.

We next spell out the promised finite matrix. At the ordered point x of (25), let E_x be the set of active annihilation edges and choose a reduced local equation d_e for each. The recurrence block is

$$B_x : p \longmapsto (p \bmod (d_e, W_e))_{e \in E_x} \in \left[\bigoplus_{e \in E_x} (R/(d_e)) \otimes (V_n/W_e)\right]^{H_x}. \tag{42}$$

The direct sum is assembled before the H_x-invariants are taken. Choose a finite R_0-module presentation

$$K_x : R_0^{m_x} \longrightarrow (R \otimes V_n)^{H_x}, \qquad \operatorname{im} K_x = \ker A_x, \tag{43}$$

and define

$$C_x = B_x K_x. \tag{44}$$

In a Walsh basis, B_x simply selects the modes not supported inside e and reduces their coefficients modulo d_e. Hence all three maps in (44) are finite analytic matrices determined directly by (20) and (34).

Proposition 4.2 (Finite stalk criterion). *At every quotient stalk,*

$$(\mathscr{N}_{n,s})_{\pi(x)} = \ker A_x, \qquad (\mathscr{J}_{n,s})_{\pi(x)} = \operatorname{im} C_x, \qquad (\mathscr{P}_{n,s})_{\pi(x)} = K_x(\ker C_x). \tag{45}$$

For prescribed boundary data η_x, a zero-transform recurrent lift exists locally exactly when the finite system

$$C_x u = \eta_x \tag{46}$$

is solvable, in which case $p_x = K_x u$ is one solution.

Proof. The first identity is Proposition 3.2. Applying B_x to $\operatorname{im} K_x = \ker A_x$ gives the second. For the third, if $p = K_x u$ and $B_x p = 0$, then $C_x u = 0$; the converse is immediate. Injectivity of K_x is not required. Equation (46) is the same pair of conditions written as

$$\begin{pmatrix} A_x \\ B_x \end{pmatrix} p = \begin{pmatrix} 0 \\ \eta_x \end{pmatrix}. \tag{47}$$

□

Theorem 4.3 (One-step lifting). *For a fixed lower zero-transform section $q_{n-2} \in \Gamma(Z_{n-2}, \mathscr{N}_{n-2,s})$, the following are equivalent.*

(i) *There is a global zero-transform q_n satisfying the recurrence with q_{n-2}.*
(ii) $G_{n,s}(q_{n-2}) \in \Gamma(Z_n, \mathscr{J}_{n,s})$.
(iii) *At every quotient stalk, (46) is solvable with $\eta_x = G_{n,s}(q_{n-2})_x$.*

When these conditions hold and q_n^* is one lift, every lift is

$$q_n^* + k_n, \qquad k_n \in \Gamma(Z_n, \mathscr{P}_{n,s}). \tag{48}$$

Proof. By Proposition 3.2, a zero-transform section is a global section of $\mathscr{N}_{n,s}$. Lemma 4.1 makes its recurrence exactly the prescribed image under $\rho_{n,s}$. This proves the necessity of (ii), and Proposition 4.2 makes (ii) equivalent to the stalkwise condition (iii).

For sufficiency, apply global sections to (41). The space Z_n is Stein and $\mathscr{P}_{n,s}$ is coherent, so Cartan B gives $H^1(Z_n, \mathscr{P}_{n,s}) = 0$. Therefore every global section of $\mathscr{J}_{n,s}$ lifts to one of $\mathscr{N}_{n,s}$. The kernel of the global map is $\Gamma(Z_n, \mathscr{P}_{n,s})$, proving (48). □

The full answer is the set of infinite paths through these one-step correspondences. Define

$$\mathcal{C}_{n,s} = \{(u, v) \in \Gamma(\mathscr{N}_{n-2,s}) \times \Gamma(\mathscr{N}_{n,s}) : \rho_{n,s}(v) = G_{n,s}(u)\} \tag{49}$$

and, for $\epsilon \in \{0, 1\}$,

$$\begin{aligned} \mathfrak{K}_s^{(\epsilon)} = \big\{ (q_{\epsilon+2m})_{m \geq 0} : \; & q_{\epsilon+2m} \in \Gamma(\mathscr{N}_{\epsilon+2m,s}), \\ & (q_{\epsilon+2m-2}, q_{\epsilon+2m}) \in \mathcal{C}_{\epsilon+2m,s} \text{ for } m \geq 1 \big\}. \end{aligned} \tag{50}$$

At level zero the transform is the identity whenever the scalar can be nonzero, so every kernel path has $q_0 = 0$.

Theorem 4.4 (Complete kernel). *For every fixed b, integer s, and fixed g as in (9),*

$$\ker T = \mathfrak{K}_s^{(0)} \oplus \mathfrak{K}_s^{(1)}, \qquad q_0 = 0. \tag{51}$$

Equivalently, at each step retain a branch precisely when $G_{n,s}(q_{n-2})$ lies in the coherent image $\mathscr{J}_{n,s}$; if it does, all next terms are one chosen lift plus $\Gamma(Z_n, \mathscr{P}_{n,s})$. Only infinite retained paths are admissible.

Proof. An admissible zero-transform family restricts at each level to a section of $\mathscr{N}_{n,s}$, and Lemma 4.1 places each successive pair in (49). Conversely, an infinite path consists of entire periodic symmetric spin sections, has zero transform at every level, and supplies an allowed remainder on every annihilation component. Particle number changes by two, so the two parities are independent. The recursive description follows from Theorem 4.3. □

Corollary 4.5 (All nonempty fibers)**.** *If p^* is one admissible lift of a target $\mathcal{K}$ in the same fixed recurrence category, then*

$$T^{-1}(\mathcal{K}) = p^* + \left(\mathfrak{K}_s^{(0)} \oplus \mathfrak{K}_s^{(1)}\right). \tag{52}$$

Proof. The difference of two lifts has zero transform, and the difference of their remainders remains tail-label independent. It therefore belongs to (51). Conversely, adding any kernel path preserves every linear admissibility condition and the transform. □

This is an exact inverse-limit parametrization, not a finite-dimensional basis. Without a growth restriction, the global section spaces appearing in (50) already contain arbitrary holomorphic functions on rapidity tori. The finite content is local: every one-step membership and homogeneous freedom is given by (44).

5. An all-level pure-A_4 target

We next construct the target in Theorem 2.4. Pillin uses an effective coupling B and

$$S_{\mathrm{P}}(\theta) = \frac{i\sinh\theta + \sin(\pi B/2)}{i\sinh\theta - \sin(\pi B/2)}. \tag{53}$$

The elementary identity

$$\frac{\tanh(\theta/2 - i\alpha)}{\tanh(\theta/2 + i\alpha)} = \frac{\sinh\theta - i\sin(2\alpha)}{\sinh\theta + i\sin(2\alpha)} \tag{54}$$

shows that (53) equals (5) under

$$B = 4b. \tag{55}$$

After the substitution $x = 2t$ in the minimal-factor integral, the two normalizations are related by

$$F_{\mathrm{P}}^{\min}(\theta) = \frac{F(\theta)}{F(i\pi)}. \tag{56}$$

Pillin's polynomial ansatz is

$$\mathcal{F}_n^{\mathrm{P}}(\boldsymbol{\theta}) = H_n Q_n(\boldsymbol{x}) \prod_{i<j} \frac{F_{\mathrm{P}}^{\min}(\theta_i - \theta_j)}{x_i + x_j}, \qquad x_j = e^{\theta_j}, \tag{57}$$

Thus the scalar factor left after removing the minimal two-particle factors is

$$\mathcal{F}_n^{\mathrm{P}} = \left(\prod_{i<j} F(\theta_i - \theta_j)\right) \mathcal{K}_n, \qquad \mathcal{K}_n = \frac{H_n}{F(i\pi)^{n(n-1)/2}} \frac{Q_n(\boldsymbol{x})}{\prod_{i<j}(x_i + x_j)}. \tag{58}$$

and its kinematic residue equation is equivalent to

$$Q_{n+2}(-x, x, \boldsymbol{x}_n) = (-1)^n D_n(x \mid \boldsymbol{x}_n) Q_n(\boldsymbol{x}_n), \tag{59}$$

where, with $\omega = e^{i\pi B/2}$,

$$D_n(x \mid \boldsymbol{x}_n) = \frac{x}{2(\omega - \omega^{-1})} \left[\prod_{j=1}^{n} (x + \omega x_j)(x - \omega^{-1} x_j) - \prod_{j=1}^{n} (x - \omega x_j)(x + \omega^{-1} x_j) \right]. \tag{60}$$

Pillin proves an all-level symmetric solution of (59); its new component at level r is

$$Q_{r,r}(x_1,\dots,x_r)=\prod_{1\le i<j\le r}(x_i+x_j) \tag{61}$$

[14, Eqs. (15), (19)].

Proposition 5.1 (Pure-A_4 family)**.** *At $b=1/4$, there is a full spin-zero admissible target $\mathcal{K}^{A_4}$ satisfying* (14)*, with all odd levels zero.*

Proof. Under (55), $b=1/4$ gives $B=1$ and $\omega=i$. In Pillin's independent-component expansion, set every free coefficient except A_4 to zero. Then

$$Q_{2,4}=0, \qquad Q_{4,4}=\prod_{1\le i<j\le 4}(x_i+x_j), \tag{62}$$

and the all-level recursion constructs $Q_{2m,4}$ for every $m\ge 2$. Define

$$\mathcal{K}^{A_4}_{2m}=C_{2m}\frac{Q_{2m,4}}{\prod_{i<j}(x_i+x_j)}, \qquad \mathcal{K}^{A_4}_{2m+1}=0, \tag{63}$$

where C_{2m} contains H_{2m} and the powers of $F(i\pi)$ from (56). Pillin's residue normalization fixes their ratios, and one overall nonzero rescaling makes $\mathcal{K}^{A_4}_4=1$.

The numerator degree is $n(n-1)/2$, equal to the pair-sum denominator degree, so (63) has spin zero. It is symmetric and separately periodic, and its only possible poles are simple pair-sum poles. Equations (59)–(60) were derived from the same kinematic residue axiom after the minimal-factor separation (57); the convention matches by (55)–(56). Thus (7) holds at every even level. Equation (62) gives $\mathcal{K}^{A_4}_0=\mathcal{K}^{A_4}_2=0$ and $\mathcal{K}^{A_4}_4=1$. □

The all-level input in Proposition 5.1 is the proved polynomial recursion, not an extrapolation from a finite prefix. We now show that its four-particle value cannot come from the labelled transform.

6. The four-particle obstruction

Throughout this section $b=1/4$, so $a=1$. Work at the ordered point

$$z^*=(-1,1,-i,i). \tag{64}$$

Only the annihilation components D_{12} and D_{34} are active. Use transverse coordinates

$$z_1=-(1+\xi), \quad z_2=1, \qquad z_3=-i(1+\eta), \quad z_4=i, \tag{65}$$

and write $d_{12}=z_1+z_2=-\xi$ and $d_{34}=z_3+z_4=-i\eta$. Split a label word as $A=(\ell_1,\ell_2)$ and $C=(\ell_3,\ell_4)$.

Let R be the ordered local ring at (64). The homogeneous recurrence condition for an upper correction q_4 is

$$q_4 \bmod d_{12}\in W_{12}, \qquad q_4 \bmod d_{34}\in W_{34}. \tag{66}$$

Lemma 6.1 (The 25-generator module)**.** *The local module defined by* (66) *is*

$$\mathscr{H}_{4,0,z^*}=R\mathbf{1}+d_{12}(R\otimes W_{34})+d_{34}(R\otimes W_{12})+d_{12}d_{34}(R\otimes\mathbb{C}^{16}). \tag{67}$$

At (64)*, the regular value of the original transform row on every displayed generator is zero.*

Proof. The two four-dimensional spaces consist of functions of disjoint label pairs, so

$$W_{12}\cap W_{34}=\mathbb{C}\mathbf{1}. \tag{68}$$

Subtract the common intersection value from the two boundary terms and divide the remaining D_{12} and D_{34} contributions by their reduced local equations. The residual section vanishes on both components and is divisible by $d_{12}d_{34}$. This gives (67).

Let e_A and e_C denote the characteristic functions of a head or tail label pair. Substitution of (65) into (19) and the coefficient row (20) gives the complete table.

generator family	multiplicity	$T_4[\text{generator}]\|_{z^*}^{\text{reg}}$
$\mathbf{1}$	1	0
$d_{12}e_C$	$C \in \{0,1\}^2$ (4)	0
$d_{34}e_A$	$A \in \{0,1\}^2$ (4)	0
$d_{12}d_{34}e_Ae_C$	$(A,C) \in \{0,1\}^4$ (16)	0

(69)

The forward residue identity with zero lower transform shows that each displayed transform is holomorphic along both active divisors. Consequently, R-linearity reduces the regular-value calculation for (67) to the 25 listed generators. For completeness, the last line has an immediate factor check. A label word that does not cut both active edges retains a factor d_{12} or d_{34} and vanishes. For each of the four proper cuts, the cross-edge product contains, up to a nonzero sign,

$$1 + \frac{a^2}{\sinh^2(i\pi/2)} = 1 + \frac{1}{i^2} = 0. \tag{70}$$

For a one-boundary generator, only the two cuts of the cleared active edge survive; their signs and cross factors are opposite at (64). The constant generator is their sum over the remaining pair. This proves all four rows of (69). There are $1 + 4 + 4 + 16 = 25$ generators. □

Lemma 6.1 says that changing p_4 while holding a lower lift fixed cannot change the regular value at z^*. We must also exclude changing the lower p_2 itself. At $b = 1/4$, write

$$u = \frac{i}{\sinh(\beta_1 - \beta_2)}. \tag{71}$$

The two-particle transform row in label order $(00, 01, 10, 11)$ is

$$(1, -1-u, -1+u, 1). \tag{72}$$

Lemma 6.2 (Symmetry-closed lower block)**.** *Let $q = (q_{00}, q_{01}, q_{10}, q_{11})^T$ be a local two-particle section with $T_2[q] = 0$ that can occur below a recurrent four-particle section. The transform equation, the shared-pair recurrence equations, and their simultaneous coordinate-label swap give*

$$M_g(u)q = 0, \tag{73}$$

where

$$M_g(u) = \begin{pmatrix} 1 & -1-u & -1+u & 1 \\ \kappa & -\kappa & -g_{00} & g_{00} \\ g_{11} & -g_{11} & -\kappa & \kappa \\ \kappa & -g_{00} & -\kappa & g_{00} \\ g_{11} & -\kappa & -g_{11} & \kappa \end{pmatrix}. \tag{74}$$

If $(g_{00}, g_{11}) \neq (\kappa, \kappa)$, its kernel is the label-constant line. If $g_{00} = g_{11} = \kappa$, the kernel is generated by

$$v^{(0)} = (1, \tfrac{1}{2}, \tfrac{1}{2}, 0)^T, \qquad v^{(1)} = (0, \tfrac{1}{2}, \tfrac{1}{2}, 1)^T. \tag{75}$$

Proof. The first row is (72). Comparing restrictions to two annihilation components that share one coordinate eliminates the tail-independent remainders and gives the next two rows; applying the simultaneous swap gives the last two. Direct 3×3 minors of (74) include

$$\pm(g_{00} - \kappa)^2, \qquad \pm(g_{11} - \kappa)^2. \tag{76}$$

The vector $(1,1,1,1)^T$ is always in the kernel. If either difference in (76) is nonzero, the rank is at least three and hence is exactly three. This includes the nonconstant locus $g_{00}g_{11} = \kappa^2$, where the unsymmetrized three-row block alone would have rank two. At the constant pair the matrix has rank two, and direct substitution shows that (75) is a basis of its kernel. □

Proposition 6.3 (No four-level lift)**.** *For every fixed diagonal pair, a recurrent spin-zero prefix with* $T_0[p_0] = T_2[p_2] = 0$ *satisfies*

$$T_4[p_4](z^*) = 0. \tag{77}$$

Consequently it cannot transform to (14).

Proof. Since T_0 is the identity at spin zero, $p_0 = 0$. First suppose the diagonal pair is nonconstant. Lemma 6.2 makes p_2 label independent, say $p_2 = f$. It has the exact recurrent upper lift $p_4 = 0$: on component e choose

$$h_{4,e}(\ell_i, \ell_j) = -g(\ell_i, \ell_j)f. \tag{78}$$

This is independent of the tail labels. Every other lift differs by a section of (67), whose transform value is zero by Lemma 6.1. Hence (77) holds.

Now let $g_{00} = g_{11} = \kappa$. A recurrent extension of a lower kernel section, modulo (67), is

$$q_{A,C}(\xi, \eta) = \kappa\big(v_A(\xi) + v_C(\eta)\big). \tag{79}$$

For $v = \alpha v^{(0)} + \delta v^{(1)}$, direct contraction with the original row gives

$$\kappa R(\xi, \eta)\big[(\alpha - \delta)(\xi) - (\alpha - \delta)(\eta)\big], \tag{80}$$

where

$$R(\xi, \eta) = \frac{4(1 + \xi)(1 + \eta)(2 + \xi + \eta)(\xi\eta + i\xi - i\eta)}{(\xi^2 + 2\xi + 2)(\eta^2 + 2\eta + 2)(\xi^2 + 2\xi + \eta^2 + 2\eta + 2)}. \tag{81}$$

Thus the regular value is zero. The apparent extra two-particle syzygy

$$v^{(\sinh)} = \left(0, \frac{i - \sinh\beta_{12}}{2}, \frac{i + \sinh\beta_{12}}{2}, 0\right)^T \tag{82}$$

would give the nonzero value $-2i\kappa$, but its shared-pair defect is $-i$; it does not satisfy (73). Hence every admissible lower change is covered by (75)–(81), and (77) again follows. □

Proof of Theorem 2.4. Proposition 5.1 supplies a full admissible target whose four-particle function is the constant one. Any hypothetical full lift would in particular give a recurrent prefix through level four. Proposition 6.3 evaluates its transform at z^* as zero, whereas $\mathcal{K}_4^{A_4}(z^*) = 1$. This contradiction is independent of the fixed diagonal pair. □

7. Low-particle controls and scope

The abstract sheaves reproduce the smallest levels directly. At $n = 0$ the kernel is zero. At $n = 1$ the row is $(1, -1)$, so

$$q_1(\beta) = ce^{s\beta}(1, 1) \tag{83}$$

is the fixed-level kernel. At $n = 2$, with $u = ia/\sinh(\beta_{12})$, the row is

$$(1, -1 - u, -1 + u, 1). \tag{84}$$

Away from its divisors it has a rank-three syzygy module. The recurrence quotient is zero at $n = 2$ because every label is an endpoint label. At $n = 3$, the quotient has label rank four, and the prescription induced by (83) is tail independent. At the first disjoint four-particle stratum,

$$\dim(W_{12} \cap W_{34}) = 1, \tag{85}$$

which is exactly the intersection used in Lemma 6.1.

Independent exact-arithmetic controls were used to check the finite algebra that supports, but does not replace, the all-level proofs.

The obstruction should not be extrapolated away from $b = 1/4$. To make this boundary concrete, write

$$t = e^{2\pi ib} = c + ia, \qquad c = \cos(2\pi b), \tag{86}$$

TABLE 1. Definition-level verification controls. The all-level ingredients are Proposition 5.1 and Theorems 4.3 and 4.4; the finite rows below check their first nontrivial instances.

Object	Exact check	Outcome
Pure-A_4 target	Componentwise polynomial recursion, permutation symmetry, and homogeneity at $n = 4, 6, 8$	All checks pass
Recurrence-zero stalk	Laurent expansion of all 25 generators in (67), including pole cancellation	All regular values vanish
Lower-lift repair	Three exact T_2 syzygies and their induced T_4 values; shared-pair defects	Values $0, 0, -2i$; defects $0, 0, -i$
Arbitrary diagonal pair	Symbolic minors of (74), the determinant-zero nonconstant locus, and 32 disjoint recurrence equations	Ranks $3, 3, 2$ in the three regimes
Complete kernel	Original rows at $n = 0, 1, 2$, recurrence quotients through $n = 4$, and the edge-swapping stabilizer	Finite presentations agree with (45)

and use the analogous point $(-1, 1, -t, t)$. For a label word ℓ and an unordered pair $e = \{i, j\}$, let $e^c = \{k, l\}$ with $k < l$, and define

$$Q_\ell(z) = \sum_{i<j} (z_i + z_j) \prod_{r\in e,\, s\in e^c} (z_r + z_s)(\ell_k - \ell_l)(z_k - z_l). \tag{87}$$

On $z_i + z_j = 0$, only the term indexed by e^c survives, and that term depends only on (ℓ_i, ℓ_j). Hence Q is a symmetric homogeneous recurrence-zero section. Multiplication by

$$H(z) = \left(\sum_{r=1}^{4} z_r^{-2}\right)^3 \tag{88}$$

makes it spin zero. Direct evaluation gives

$$T_4[HQ](-1, 1, -t, t) = 4096\, ia^3c^4. \tag{89}$$

This is nonzero whenever $c \neq 0$. Thus the constant-germ cokernel used in Section 6 disappears at this same locus for generic coupling. Equation (89) neither constructs an all-level lift nor proves generic surjectivity; it only rules out reusing the $b = 1/4$ point-value proof unchanged.

There is a second scope boundary. Cartan B supplies holomorphic lifts but no growth estimates. Consequently Theorem 4.4 answers the unrestricted conditions in Definition 2.2; it does not by itself classify the polynomial-growth subclass used for correlation estimates in [10]. Likewise, the quotient (35) encodes the reduced-divisor value recurrence. A jet-level or nonreduced recurrence would require a different quotient sheaf.

8. Conclusion

The Sinh–Gordon K-transform is not a universal parametrization of the unrestricted bootstrap families. At $b = 1/4$, the all-level pure-A_4 target has an empty fiber for every fixed diagonal recurrence pair. The failure is visible at four particles, but it is not a truncated-target artifact: Pillin's recursion supplies the full admissible family.

For every fixed coupling and fixed recurrence parameters, all integer-spin fibers nevertheless have a complete common form. The original transform row and the tail-independence quotient give finite stalk matrices A_x, B_x, and $C_x = B_xK_x$. Cartan B turns stalkwise image membership into a global one-step lift, and the two parity inverse limits give the exact kernel. Every target fiber is therefore either empty or an affine translate of this explicitly presented path space.

The remaining image problem is parameter dependent. In particular, (89) shows that the exceptional four-particle value obstruction does not persist at the same locus for generic b. Determining the image there requires a different stratum, higher normal jets, or an all-level compatibility obstruction. Adding a growth condition would pose a separate analytic lifting problem because the Stein-space argument used here does not control growth.

Data and code availability

No empirical data were used, and no step of any proof above is computer assisted. The all-level results rest on Proposition 5.1 and Theorems 4.3 and 4.4, proved analytically here; the finite algebra recorded in Table 1 is stated and derived in the text, at (67), (74) and (45), and each entry can be recomputed by hand from the displayed coefficient row (20). Exact rational and symbolic arithmetic was run independently while preparing the paper, and it reproduced those entries; because it certifies a transcription rather than an argument, nothing above depends on it.

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